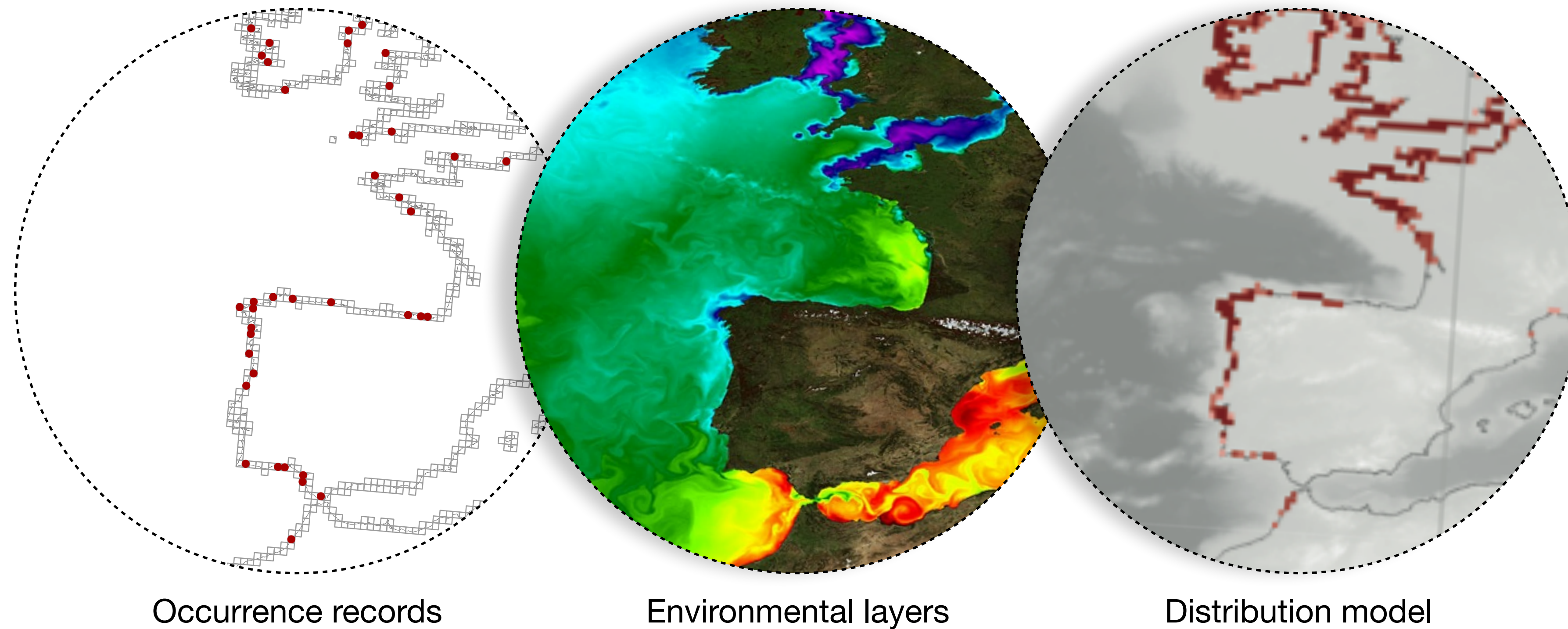


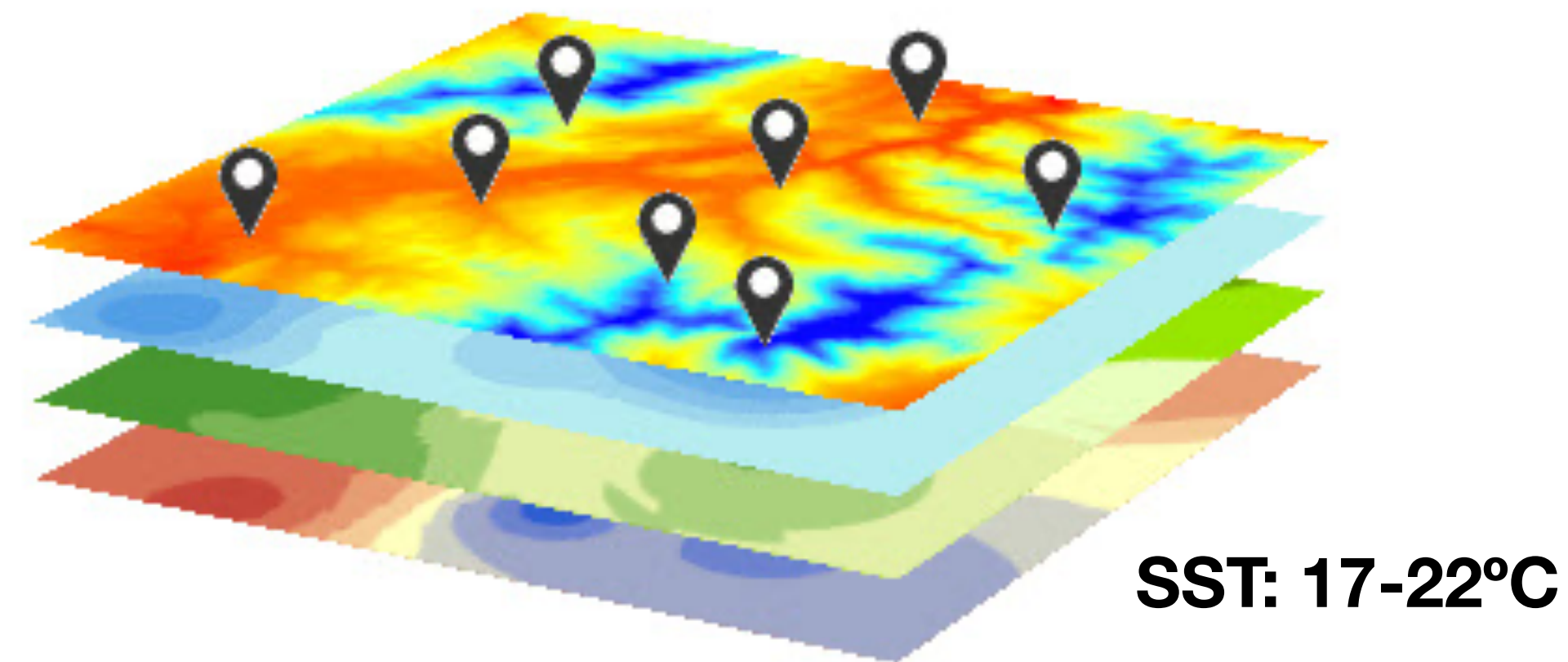
Biological and environmental data for SDM

Modeling the distribution of marine biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal



SDMs rely on species occurrence records (where are the species found) and environmental data layers (what are the conditions across the study area).



Occurrence records for modeling

Allow building an SDM and answer the question of **why a species occurs in a given area**.

Occurrence records need to reference, at least, a **particular taxon**, at a **particular place**, and a **particular time**.

Geographic coordinates are key, as they allow **linking occurrences records with environmental conditions at sites**.

Metadata can describe the source of record (sampling method), means of identification (e.g., visual census), and spatial precision of record (maximum error).



Occurrence data for SDM : one species example

Data

Metadata

Lon	Lat	Date	Depth	Species	Common name	Behaviour	Observer
0,61287	39,58955	10/08/12	22	Caretta caretta	Loggerhead sea turtle	Swimming	John Doe
0,83999	39,77417	03/03/12	12	Caretta caretta	Loggerhead sea turtle	Swimming	John Doe
3,18192	42,40479	14/08/11	1	Caretta caretta	Loggerhead sea turtle	Eating	Unknown
-4,86395	55,24295	01/01/04	0	Caretta caretta	Loggerhead sea turtle	Swimming	Unknown
-5,89297	57,06977	03/03/12	0	Caretta caretta	Loggerhead sea turtle	Sleeping	John Doe
34,9082	32,55822	22/12/02	23	Caretta caretta	Loggerhead sea turtle	Sleeping	John Doe
0,61287	39,58955	14/08/11	12	Caretta caretta	Loggerhead sea turtle	Swimming	John Doe
0,83999	39,77417	03/03/12	2	Caretta caretta	Loggerhead sea turtle	Sleeping	John Doe
3,18192	42,40479	22/12/02	5	Caretta caretta	Loggerhead sea turtle	Eating	Unknown
-4,86395	55,24295	03/03/12	7	Caretta caretta	Loggerhead sea turtle	Swimming	Unknown
-5,89297	57,06977	01/01/04	3	Caretta caretta	Loggerhead sea turtle	Eating	Unknown

Occurrence records: availability

Where is the data available for modelling?

Global Biodiversity information centres;

Literature;

Reference collections (e.g. museums and herbaria);

Citizen science initiatives.

Occurrence records: Global Biodiversity information centres

(i.e., databases linked via a common access portal).

Global Biodiversity Information Facility (GBIF):

Occurrence records: over 1.9 billion

Ocean Biodiversity Information System (OBIS):

Occurrence records: over 50 million

National Biodiversity Network (NBN) Atlas (UK):

Occurrence records: over 200 million

Atlas of Living Australia (ALA):

Occurrence records: over 95 million

Occurrence records: Biological reference collections

Preserved specimens / photographs maintained in **museums, universities, botanic gardens**, and similar **institutions**.

A major tools for basic investigation and assessment of biodiversity.

~2.5 billion specimens in preserved collections.

Macroalgae Herbarium Portal (www.macroalgae.org):

Occurrence records: 1 million records.

Occurrence records: Literature

Literature is useful to compile biodiversity records.

Peer-reviewed publications are preferred, even when out of the scope of our work.

The screenshot displays the Web of Science search results interface. At the top, navigation links include Web of Science™, InCites®, Journal Citation Reports®, Essential Science Indicators™, and EndNote®, along with Sign In, Help, and English options. The main header features the Web of Science™ logo and the Thomson Reuters™ logo. Below the header, a 'Back to Search' button is visible on the left, and 'My Tools', 'Search History', and 'Marked List' links are on the right. The search results section shows 'Results: 1,766 (from All Databases)' and 'You searched for: TOPIC: ("Caretta caretta") ...More'. The 'Refine Results' section on the left includes a search bar and a list of research domains: Science Technology, Social Sciences, and Arts Humanities. The main results area lists three publications, each with a title, author, journal information, and a 'View Abstract' button. The first publication is 'Stable isotopic comparison between loggerhead sea turtle tissues' by Vander Zanden, Hannah B; Tucker, Anton D; Bolten, Alan B; et al., published in 'Rapid communications in mass spectrometry : RCM' Volume 28 Issue 19 Pages: 2059-64 Published: 2014-Oct-15. The second publication is 'The geomagnetic environment in which sea turtle eggs incubate affects subsequent magnetic navigation behaviour of hatchlings' by Fuxjager, Matthew J; Davidoff, Kyla R; Mangiamele, Lisa A; et al., published in 'Proceedings. Biological sciences / The Royal Society' Volume 281 Issue: 1791 Published: 2014-Sep-22. The third publication is 'Levels and profiles of POPs (organochlorine pesticides, PCBs, and PAHs) in free-ranging common bottlenose dolphins of the Canary Islands, Spain' by Garcia-Alvarez, Natalia; Martin, Vidal; Fernandez, Antonio; et al., published in 'SCIENCE OF THE TOTAL ENVIRONMENT' Volume: 493 Pages: 22-31 Published: SEP 15 2014. Each publication also shows 'Times Cited: 0 (from All Databases)'.

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Refine Results

Search within results for...

Databases

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- ☐ SOCIAL SCIENCES
- ☐ ARTS HUMANITIES

Sort by: Publication Date -- newest to oldest

Page 1 of 177

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1. **Stable isotopic comparison between loggerhead sea turtle tissues.**
By: Vander Zanden, Hannah B; Tucker, Anton D; Bolten, Alan B; et al.
Rapid communications in mass spectrometry : RCM Volume: 28 Issue: 19 Pages: 2059-64 Published: 2014-Oct-15
Services View Abstract Times Cited: 0 (from All Databases)

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SCIENCE OF THE TOTAL ENVIRONMENT Volume: 493 Pages: 22-31 Published: SEP 15 2014
Services View Abstract Times Cited: 0 (from All Databases)

Occurrence records: Literature

Literature is useful to compile biodiversity records.

Peer-reviewed publications are preferred, even when out of the scope of our work.

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Back to Search

Results: 1,766
(from All Databases)

You searched for: TOPIC: ("Caretta caretta") ...More

Refine Results

Search within results for...

Databases

Research Domains

- ☐ SCIENCE TECHNOLOGY
- ☐ SOCIAL SCIENCES
- ☐ ARTS HUMANITIES

Sort by: Publication

1. Stable isotope analysis of loggerhead sea turtles (*Caretta caretta*) from the Cape Canaveral National Seashore (CNS), on the east coast of Florida in 2004 (n = 13), and Casey Key (CK), on the west coast of Florida, in 2011 (n = 20) between the months of May and July. Prior to collection, the sampling site was cleaned with isopropyl alcohol. Scute and skin were collected with 5 or 6 mm biopsy punches. The skin samples contained both epidermal and dermal layers.

2. The geomorphology of the Cape Canaveral National Seashore (CNS), on the east coast of Florida, in 2004 (n = 13), and Casey Key (CK), on the west coast of Florida, in 2011 (n = 20) between the months of May and July. Prior to collection, the sampling site was cleaned with isopropyl alcohol. Scute and skin were collected with 5 or 6 mm biopsy punches. The skin samples contained both epidermal and dermal layers.

3. Levels and profiles of POPs (organochlorine pesticides, PCBs, and PAHs) in free-ranging common bottlenose dolphins of the Canary Islands, Spain

By: Garcia-Alvarez, Natalia; Martin, Vidal; Fernandez, Antonio; et al.

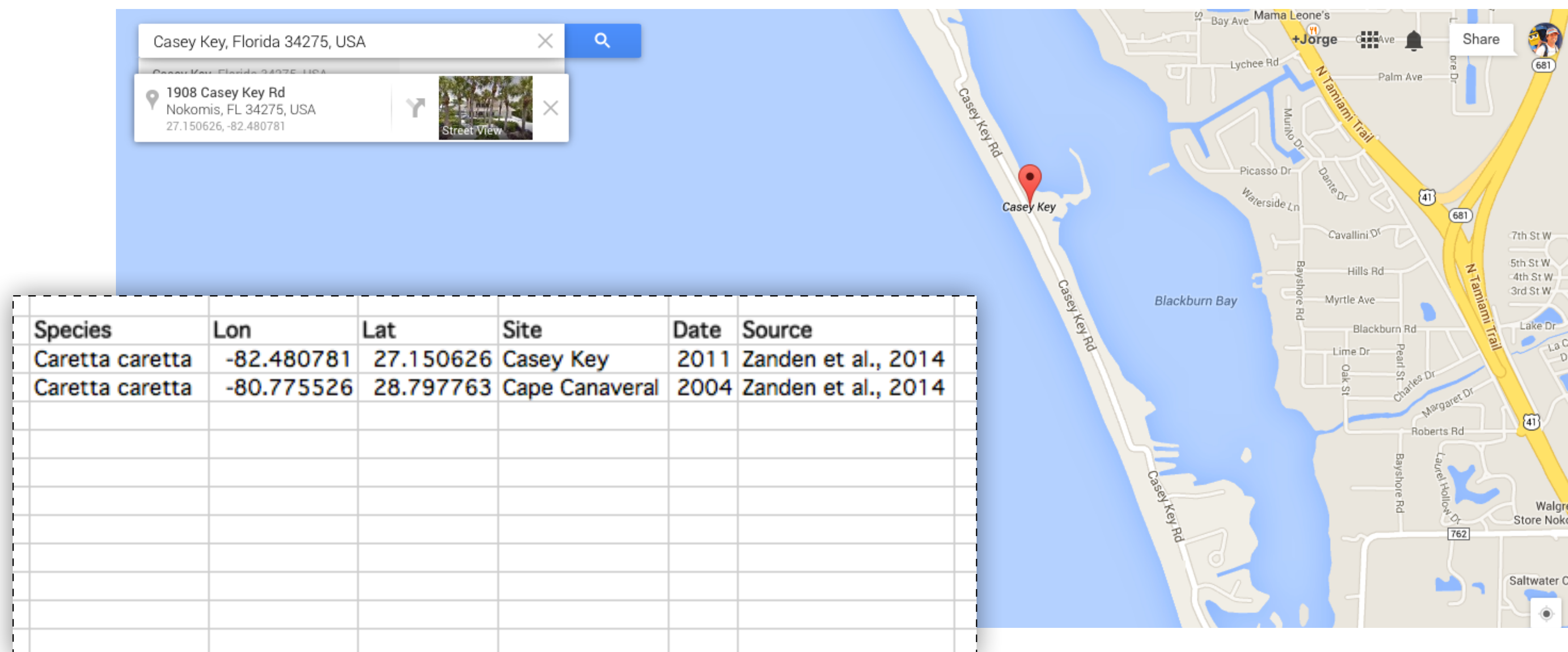
SCIENCE OF THE TOTAL ENVIRONMENT Volume: 493 Pages: 22-31 Published: SEP 15 2014

Times Cited: 0
(from All Databases)

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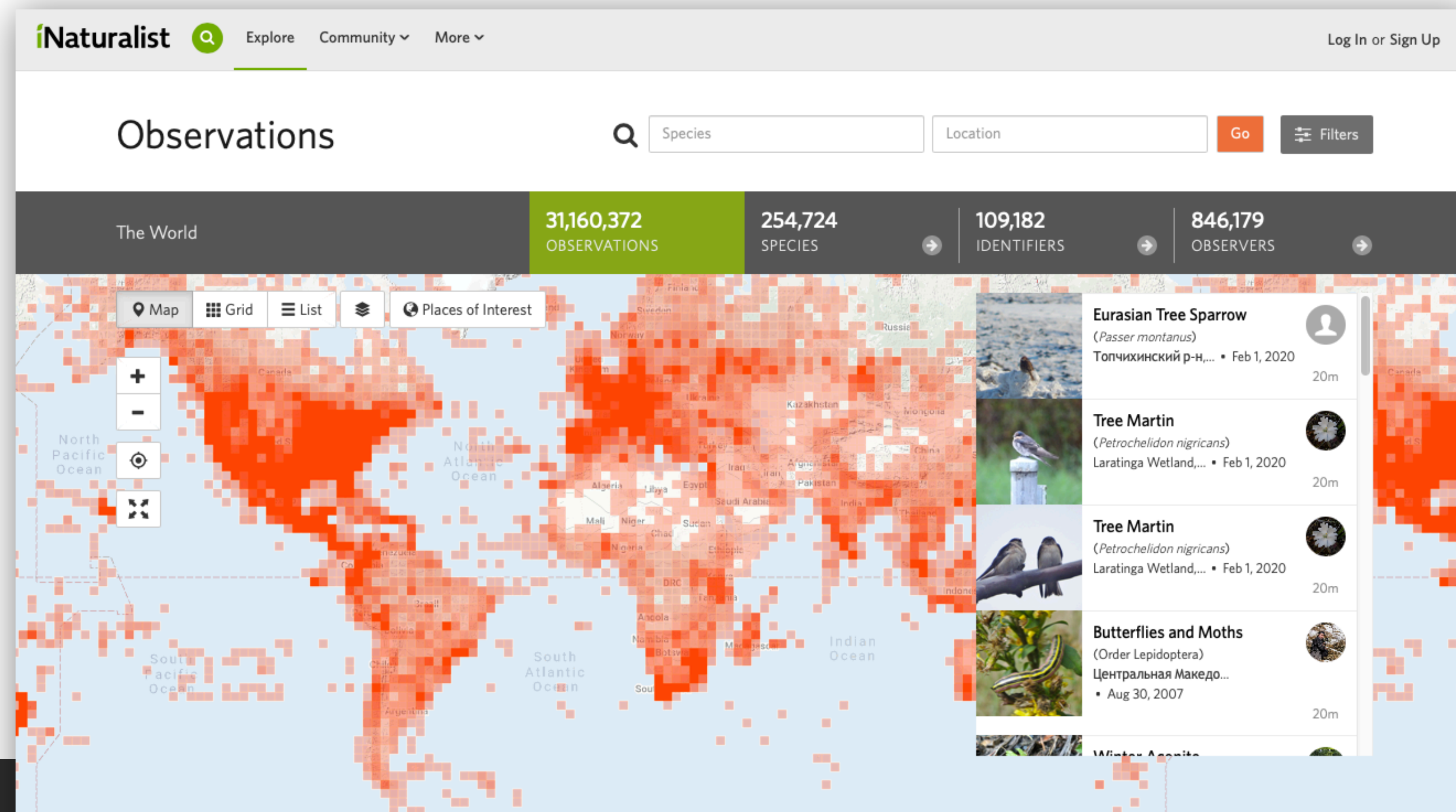
Peer-reviewed publications are preferred, even when out of the scope of our work.



Occurrence records: Citizens can help

Sharing biodiversity records / pictures with scientists.

<http://www.inaturalist.com>



Occurrence records: Citizens can help

Sharing marine forest (kelps, seagrasses, etc.) pictures with scientists.

<http://www.marineforests.com>

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GLOBAL MAPADD RECORDS

Marine Forests Data

The data provided are available as open source for everyone to use.

Scientific Name

Region (locality, district, country)

Date : from

Date : to

☒ Pictures

☐ Literature

☐ Herbarium

☐ Curated

Sort by

Sorting

QUERY

6186 records (633 species) with pictures

DOWNLOADGALLERY VIEWMAP VIEW

Species	Site	Observed	Added	User
 <i>Fucus vesiculosus</i> Linnaeus	Portugal	Jan. 5, 2020	Jan. 8, 2020, 12:40 p.m.	Jan Verbeek
				

Occurrence records: issues

“Garbage In, Garbage Out”: model outputs are only as good as the input data.

The quality of several sources has been questioned.

There are **inaccuracies** and **bias** in occurrence records.

Inaccurate records

1. Inaccurate taxonomy of species

Manual **curation and expert judgment is needed** prior to modeling.

One must know the overall distribution of the modeled taxon to properly judge occurrence records.

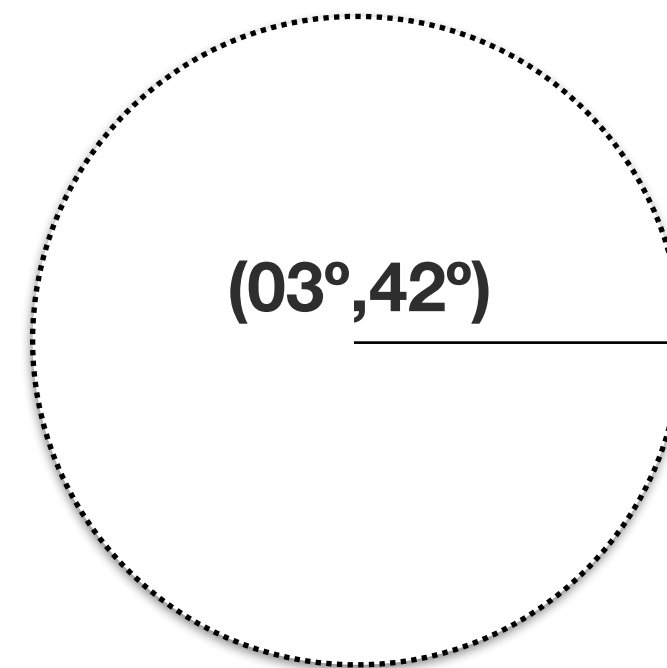
2. Inaccurate coordinates

Coarse resolution data has little meaning on where record were made.

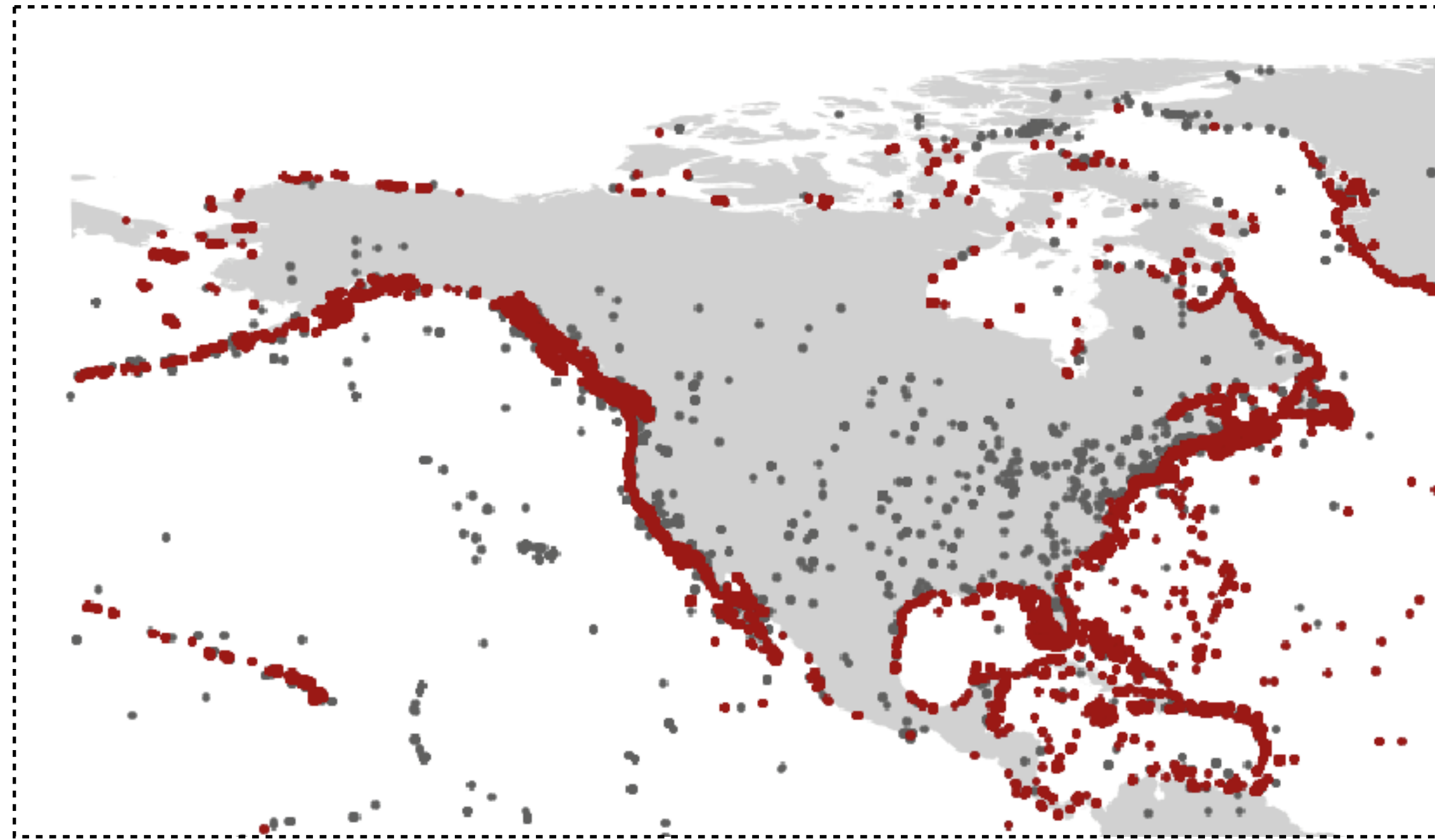
e.g.,

Natural history specimens may have very general **locality information**.

Brazil, Rio de Janeiro: locality descriptor refers to the **city** or the **state**? Also, in 2022 it is **a broader city** than it was in 1880. Also, how far outside of the limits of Rio de Janeiro the specimen could have been collected and still be labeled as “Rio de Janeiro”.



“Longitude: 3° and Latitude 42°”: A record with an accuracy at the scales of 1 degree has a potential error of ~111 kilometres.



Inaccurate records

Marine records over land (and vice versa)

Can also arise from coordinate imprecision, flipped Lon-Lat or wrong source information (the museum location instead of records).

Can be easily dealt with.

Biased records

Detectability

Some species more easily detected.

The **more mobile or cryptic the species, the greater is the detection problem.**

Raises the change of **false absences** when characterizing biodiversity data, which can lead to wrong estimates of unsuitable habitats.

This sort of bias **rarefies biodiversity without introducing systematic biases, because the effect is consistent across range.**

Biased records

Distribution effort

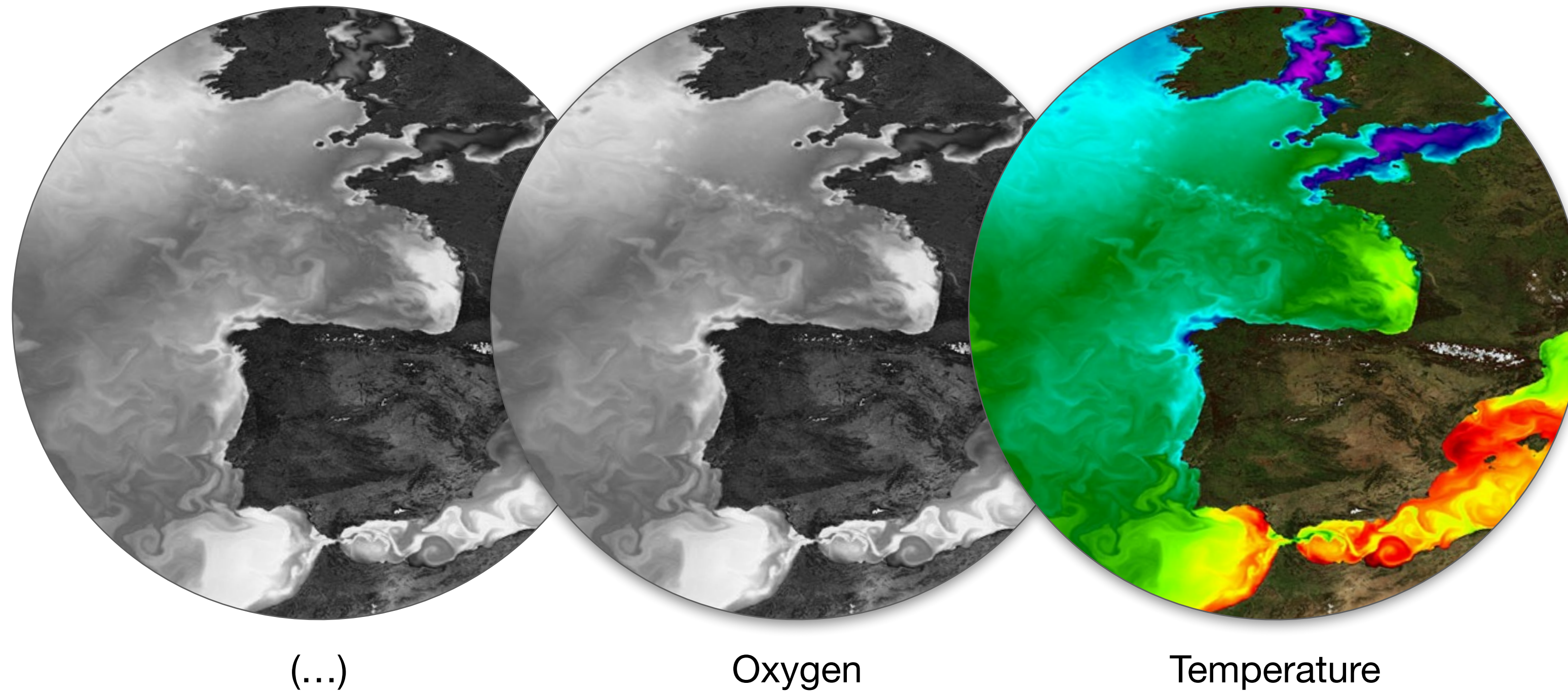
Concentrate records, providing a skewed representation of biodiversity.

1. Sampling **near roadways, rivers**, areas of high human **population density** (common in terrestrial ecology).
2. The presence of a particular **institution** or **investigator in a given area**, or the location of a popular **field station**.
3. Particular organizations or **scientists** not **willing to share data**;
4. **Inaccessibility to particular regions, e.g., deep waters**, creates a truncated distributed towards shallow waters.

Biodiversity data cleaning

Particularly important for data sourced from warehouses such as GBIF.

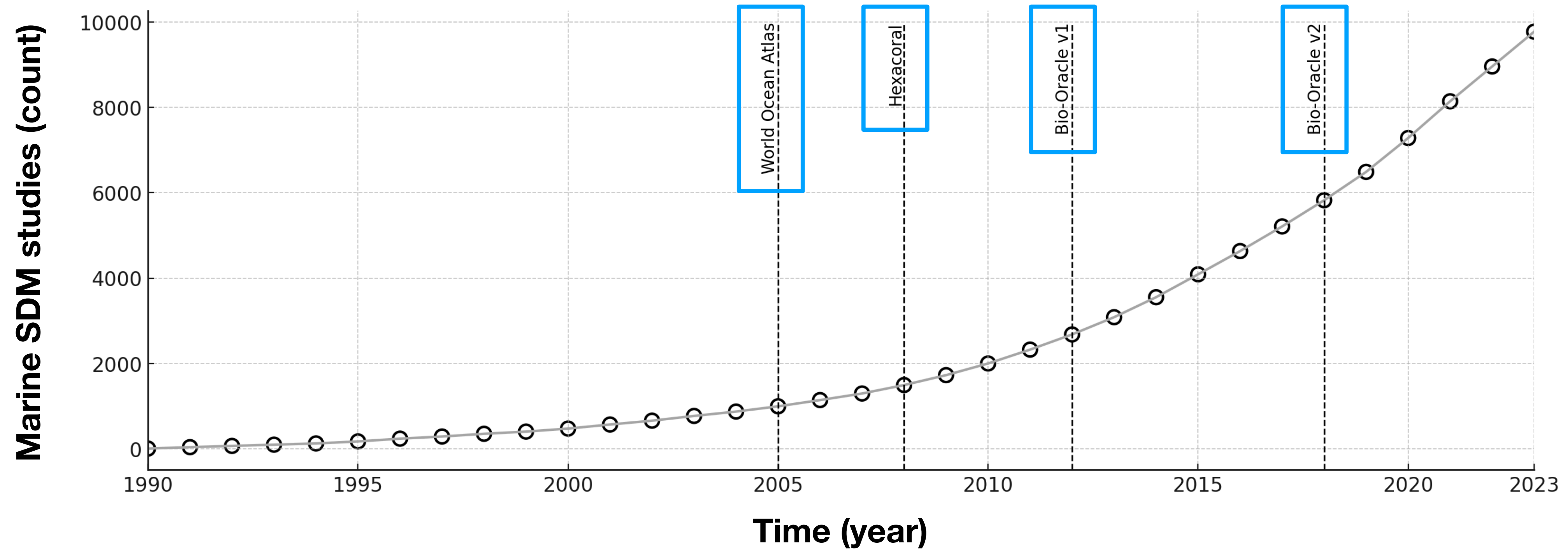
1. Remove records with NA coordinates;
2. Remove records on land or outside known distributions (depth also);
3. Remove duplicate records.



Environmental data

Allows **linking biodiversity data with their extant conditions.**

Environmental predictors typically in the form of digital raster layers.



Increase use of marine SDM in climate change projections owing to pioneer initiatives ** that provide **open-source GIS data layers with uniform resolution, global coverage**, and variables with **biological significance**.

Received: 30 July 2023 | Revised: 24 January 2024 | Accepted: 24 January 2024

DOI: 10.1111/geb.13813

DATA ARTICLE

Global Ecology
and Biogeography

A Journal of
Macroeology

WILEY

Bio-ORACLE v3.0. Pushing marine data layers to the CMIP6 Earth System Models of climate change research

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Bart Vanhoorne³  | Lennert Tyberghein³  | Ester A. Serrão¹  |
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⁵School of BioSciences, University of Melbourne, Parkville, Victoria, Australia

⁶Department of Biology, Phycology Research Group, Ghent University, Ghent, Belgium

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Funding information

HORIZON EUROPE Framework Programme, Grant/Award Number: HORIZON-CL6-2021-BIODIV-01-12; Research Foundation - Flanders

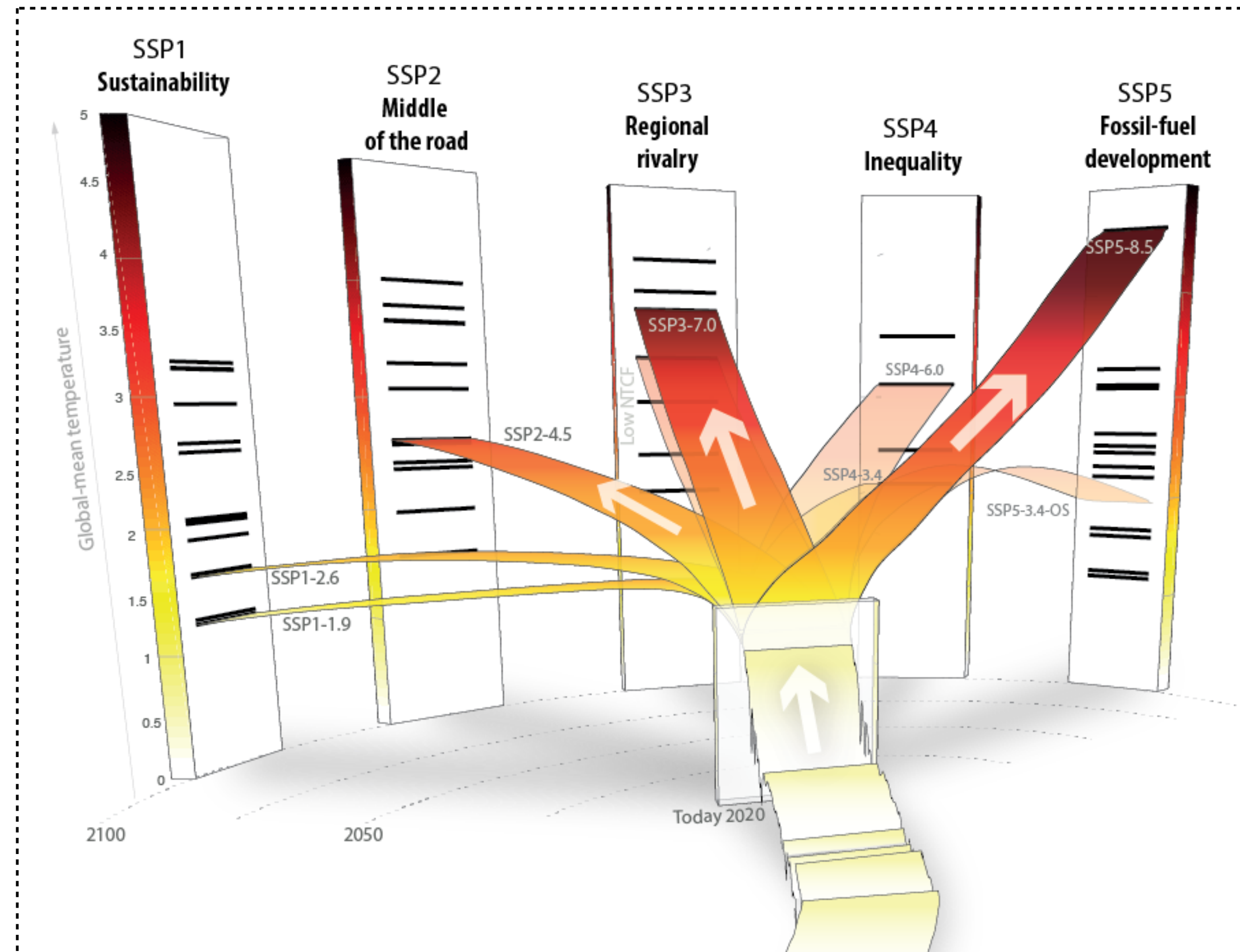
Abstract

Motivation: Impacts of climate change on marine biodiversity are often projected with species distribution modelling using standardized data layers representing physical, chemical and biological conditions of the global ocean. Yet, the available data layers (1) have not been updated to incorporate data of the Sixth Phase of the Coupled Model Intercomparison Project (CMIP6), which comprise the Shared Socioeconomic Pathway (SSP) scenarios; (2) consider a limited number of Earth System Models (ESMs).

Bio-Oracle improvements

Layers for future conditions **consider data from the 6th phase of the Coupled Model Inter-comparison Project.**

CMIP6 (1) **is more accurate in capturing the spatial and temporal variability of oceanographic conditions** and (2) **outputs the SSP scenarios**, which are the new narratives of global change.



The new **SSP** use up-to-date socioeconomic assumptions and encompass the Paris Agreement expectations, allowing the development of projections aligned with international climate policies.

Bio-Oracle improvements

Layers for future conditions comprise numerous Earth System Models.

ESM are simulations that resolve global atmospheric and oceanographic conditions across space and time (e.g., end-of-century SST in Culatra island).

ESM have inherent uncertainties.

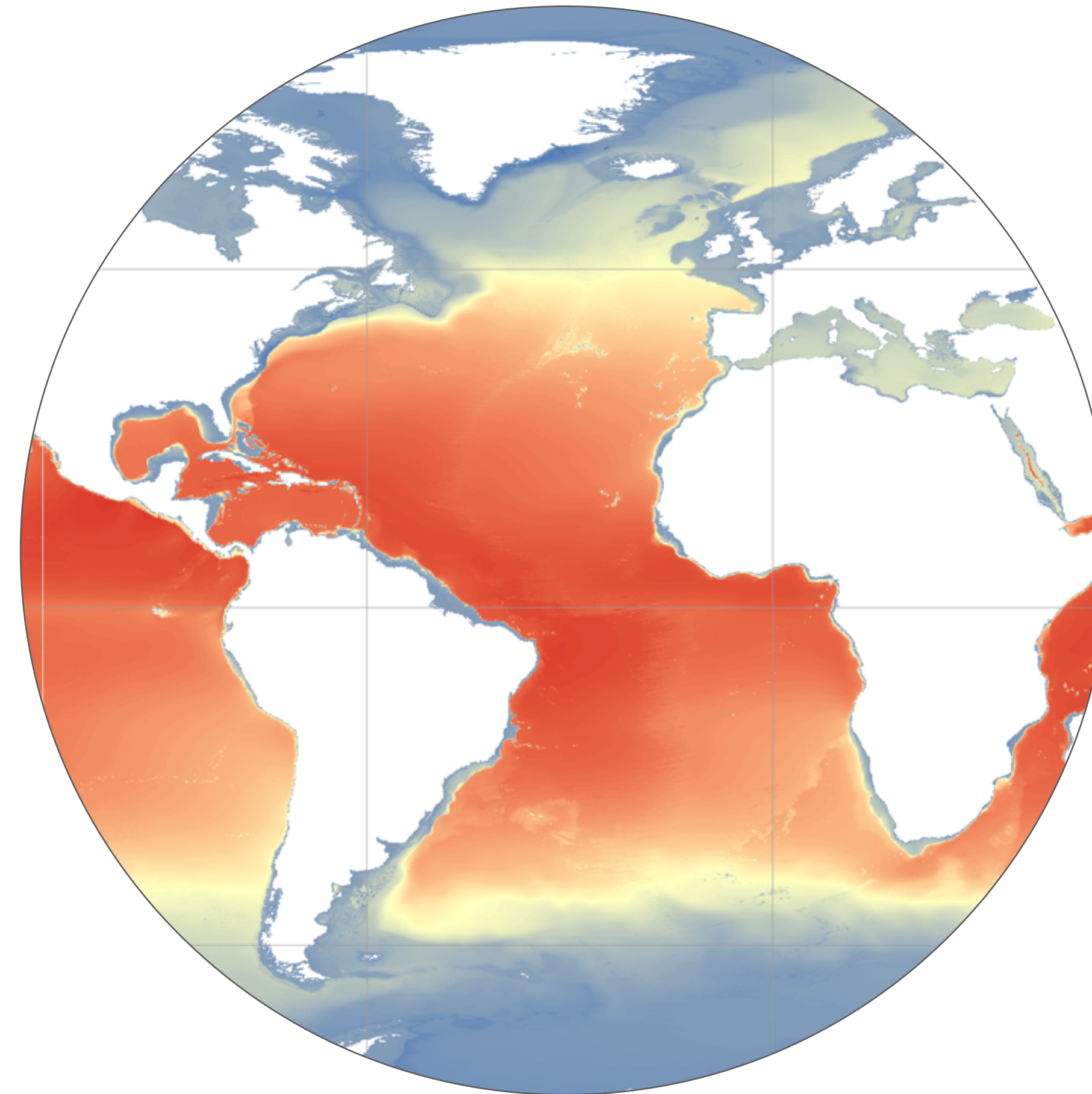
The multi-model ensemble (e.g., averaging different models) to account for inter-ESM variability. Previous available datasets limited to 2 ESMs (from the dozens available), limiting the potential to capture the range of climate change.

Bio-Oracle improvements

Layers for future conditions consider biologically meaningful variables beyond temperature, salinity and sea ice conditions.

Oxygen, primary productivity and pH (among others) are limiting factors for biodiversity, and are expected to change in the future.

Failing to consider these variables in climate change projections might misinterpret the exposure of biodiversity to future detrimental conditions.



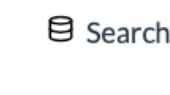
-0.5 28.8

Difference between surface and bottom temperature

All layers also provided for bottom conditions, to better model and project changes of benthic species (e.g., deep corals, or demersal fish).

		Present	SSP1-1.9	SSP1-2.9	SSP2-4.5	SSP3-7.0	SSP4-6.0	SSP5-8.5
		2 (2000-2020)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)
1	Temperature							
2	Salinity							
3	Sea Ice Cover							
4	Sea Ice Thickness							
5	Sea Water Velocity							
6	Mixed Layer Depth							
7	Diffuse Attenuation Coefficient							
8	PAR							
9	PAR at Bottom							
10	Oxygen							
11	pH							
12	Iron							
13	Phosphate							
14	Nitrate							
15	Silicate							
16	Primary Production							
17	Phytoplankton							
18	Chlorophyll							

Extension of Bio-ORACLE: 18 essential variables provided for the surface and along the seafloor, **from 2000 to 2100, per decade, at 0.05° resolution, covering the span of SSPs**, by ensembling 10 ESMs of CMIP6.


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From

1992-01-01

To

2022-11-06

Parameters

Protocols

Only the whole selected time range

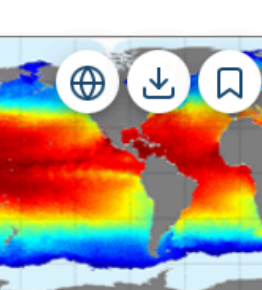
Only with depth level

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Ocean Monitoring Indicator catalogue

There is 54 ocean products corresponding to your criteria

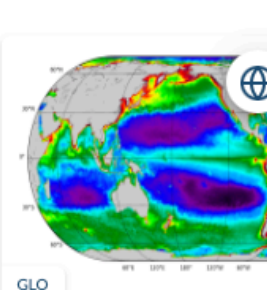


Global Ocean 1/12° Physics Analysis And Forecast Updated Daily

GLOBAL_ANALYSIS_FORECAST_PHY_001_024

T bottomT S SSH UV UV MLD SIC SIT SIUV

From 2019-01-01 To Present

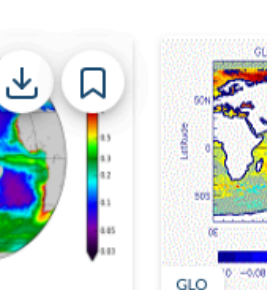


Global Ocean Biogeochemistry Analysis And Forecast

GLOBAL_ANALYSIS_FORECAST_BIO_001_028

CHL PHYC O2 NO3 PO4 SI FE SPCO2 PH PP

From 2019-05-04 To Present

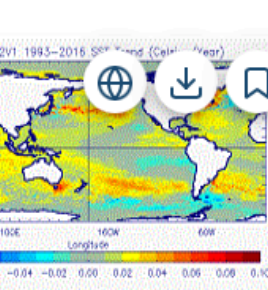


Global Ocean Physics Reanalysis

GLOBAL_MULTIYEAR_PHY_001_030

T bottomT S SSH UV MLD SIC SIT SIUV

From 1993-01-01 To 2020-05-31

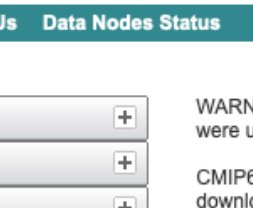


Global Ocean Ensemble Physics Reanalysis - Low Resolution

GLOBAL_REANALYSIS_PHY_001_026

T S UV SIC SIT

From 1993-01-01 To 2019-12-15



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Product

Source ID

Institution ID

Source Type

Nominal Resolution

Experiment ID

Sub-Experiment

Variant Label

Grid Label

Table ID

Frequency

Realm

Variable

CF Standard Name

Data Node

WARNING: Not all models include a variant "r1i1p1f1", and across models, identical values of variant_label do not imply identical variants! To learn which forcing datasets were used in each variant, please check modeling group publications and documentation provided through ES-DOC.

CMIP6 project data downloads are unrestricted. Downloads should be performed with the -s option to a wget script without the need to login. When using this method for download, ensure you are not using additional options, eg. -s and -H should never be combined.

For more information about CMIP6 data please consult this guide: <https://pcmdi.llnl.gov/CMIP6/Guide/dataUsers.html>

Please try our updated ESGF web application (named "Metagrid"), now undergoing beta testing. For this test release we are reaching out for help from users in the community to report any issues they encounter with the application. The beta-test web application can be found at the following site: <https://aims2.llnl.gov/metagrid/>
Please see the following page for more information including a FAQ: <https://esgf.github.io/esgf-user-support/metagrid.html>

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Total Number of Results: 5942154

-1- 2 3 4 5 6 Next >>

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1. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.Amon.wap.gn

Data Node: crd-esgf-drc.ec.gc.ca

Version: 20190429

Total Number of Files (for all variables): 1

Full Dataset Services: [Show Metadata](#) [List Files](#) [WGET Script](#) [LAS](#) [Show Citation](#) [PID](#) [Further Info](#)

2. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.day.clt.gn

Data Node: crd-esgf-drc.ec.gc.ca

Version: 20190429

Total Number of Files (for all variables): 1

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3. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.day.hfls.gn

Data Node: crd-esgf-drc.ec.gc.ca

Version: 20190429

Total Number of Files (for all variables): 1

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4. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.day.hfss.gn

Data Node: crd-esgf-drc.ec.gc.ca

Version: 20190429

Total Number of Files (for all variables): 1

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5. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.day.hursmax.gn

Data Node: crd-esgf-drc.ec.gc.ca

Version: 20190429

Total Number of Files (for all variables): 1

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6. CMIP6.ScenarioMIP.CCCma.CanESM5.ssp126.r12i1p2f1.day.hursmin.gn

Data Node: crd-esgf-drc.ec.gc.ca

Version: 20190429

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Data from Copernicus Marine Service and World Climate Research Programme



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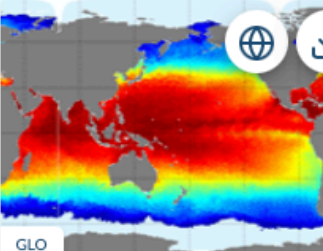
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All Area

☐ Only the whole selected time



GLO

Global Ocean 1/12° Physics Analysis Forecast Updated Daily

GLOBAL_ANALYSIS_FORECAST_PHY_001

T bottomT S SSH UV MLD SIC SIT SIUV

From 2019-01-01 To Present

WCRP



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which forcing datasets

using this method for

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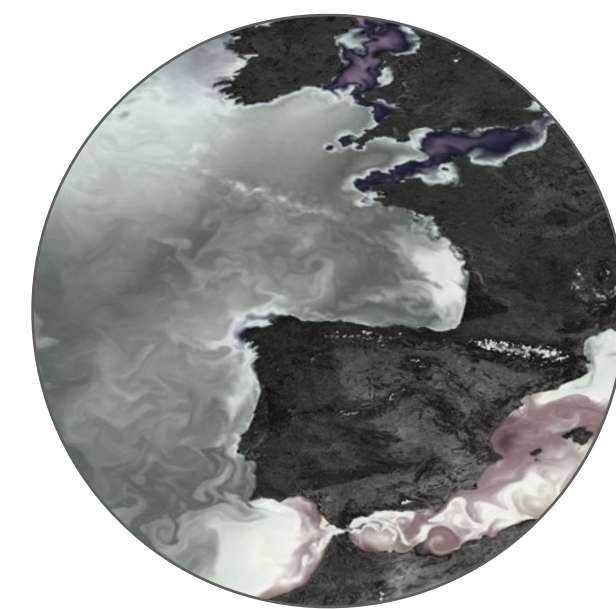
[h Options \]](#)



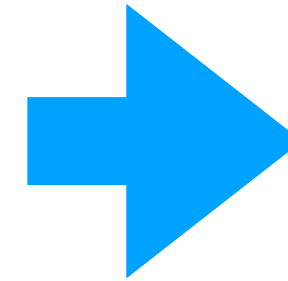
This helps evaluate climate models, leads to improvements in the model simulations and provides a better understanding of past, present and future climates.

Data f

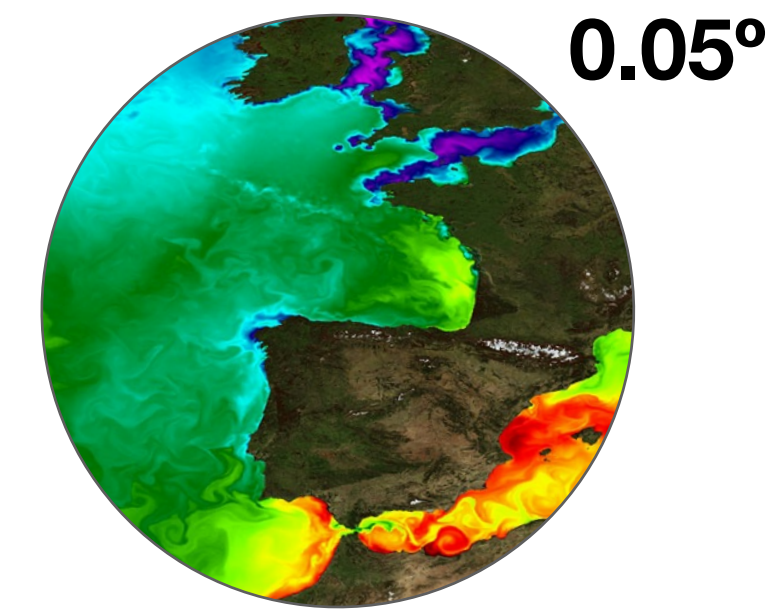
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Subsetting data and
resampling to BO resolution

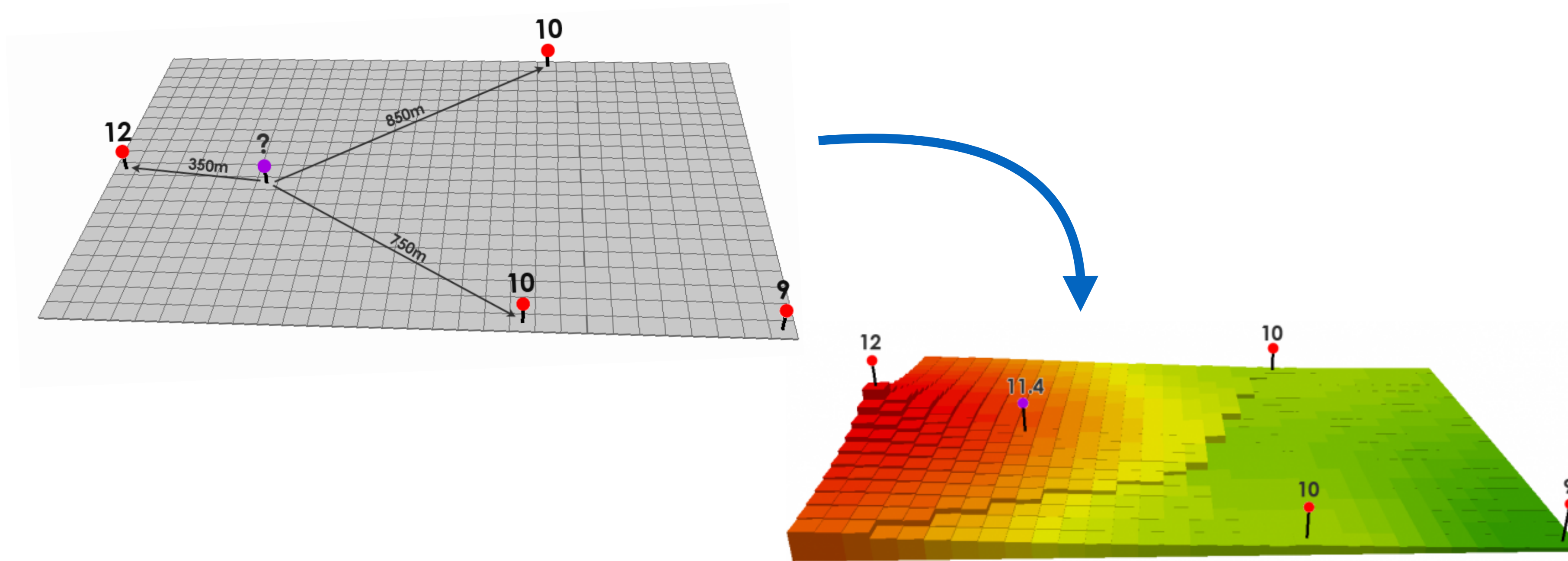


BO Layer
2010-2020

For present-day conditions, layers are developed by subsetting data to the desired decade and resampling (interpolating) to the resolution of BO (0.05°).

e.g.,

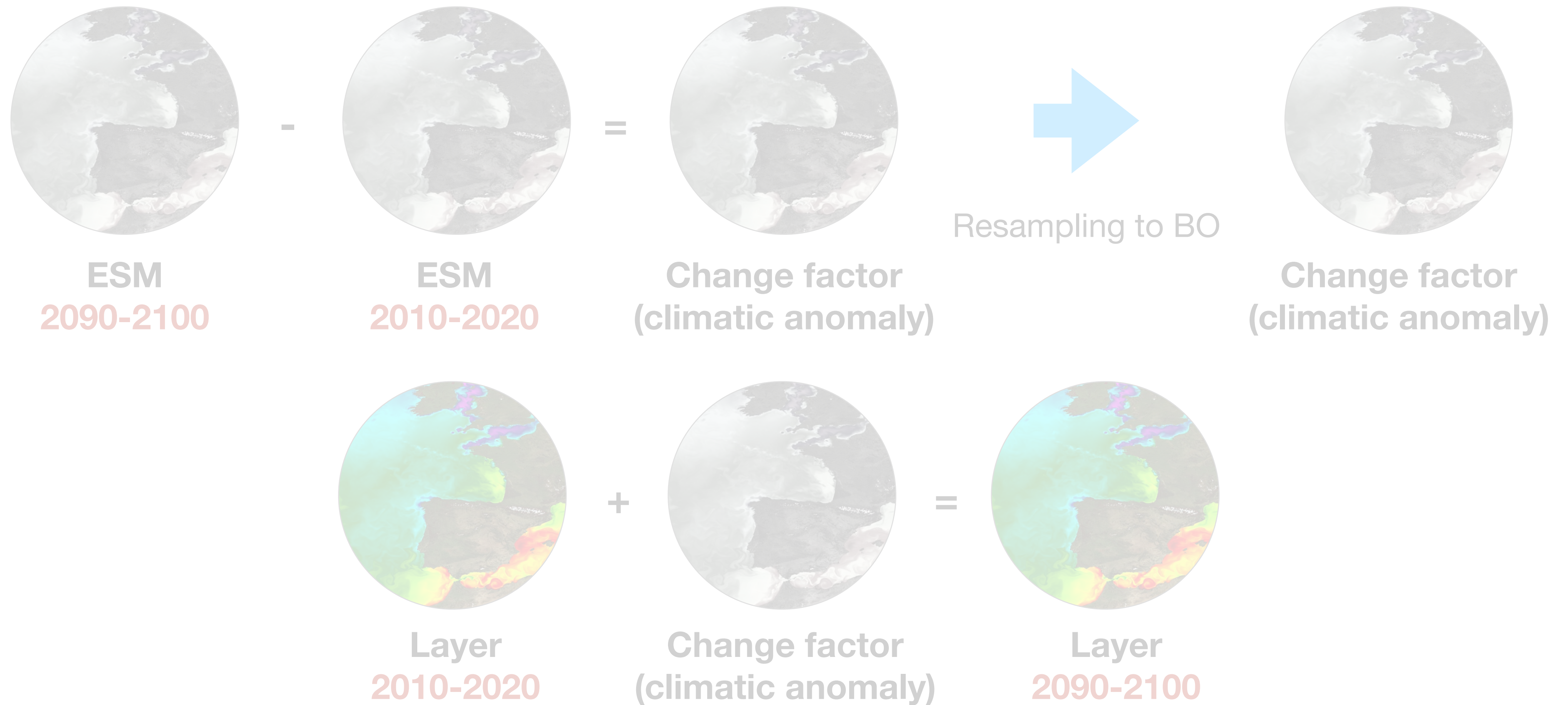
layer 1: Global maximum sea surface temperature of 2010-2020



Interpolation

Data are collected from numerous sources (e.g., buoys, loggers in the field) and their information are spatially interpolated to infer the distribution of the environmental gradients for the whole region - **estimates unknown points**.

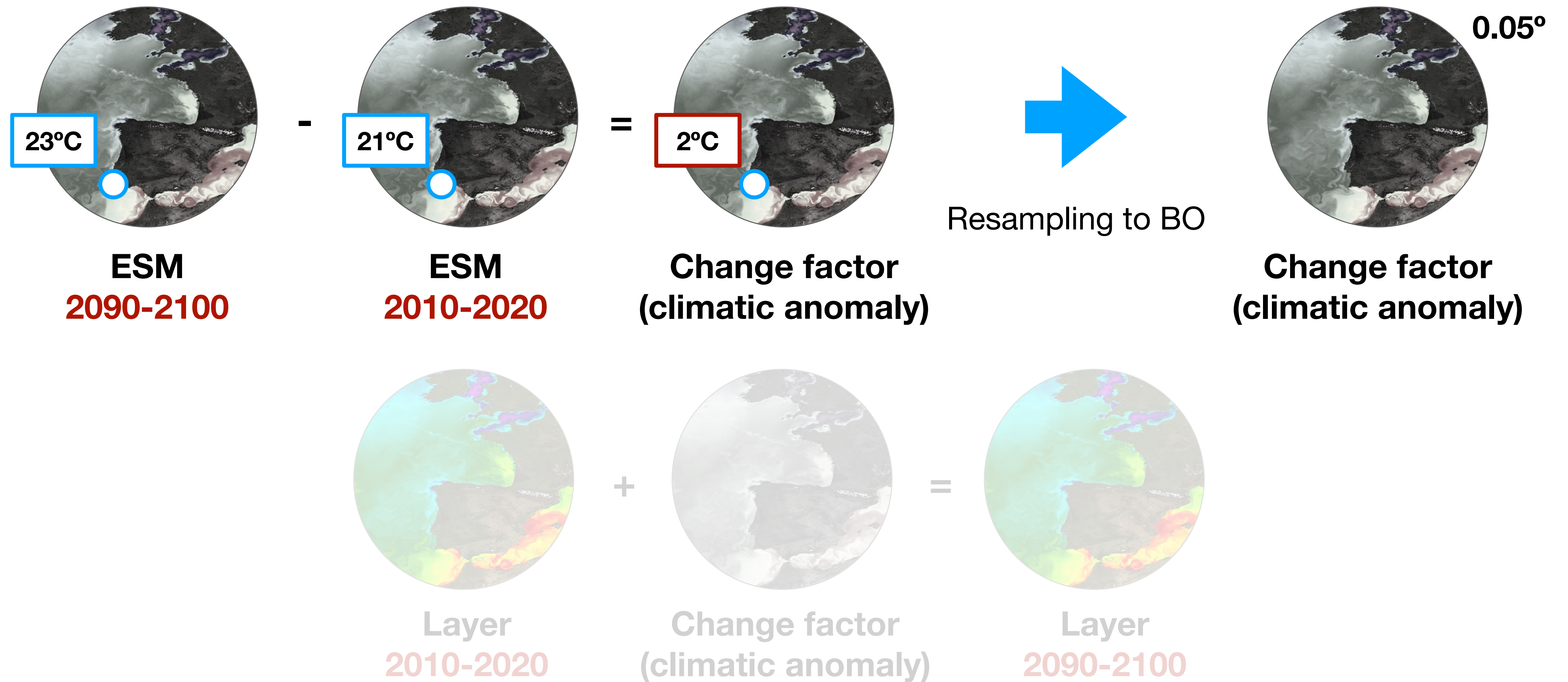
e.g., Inverse Distance Weighting algorithm.



For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

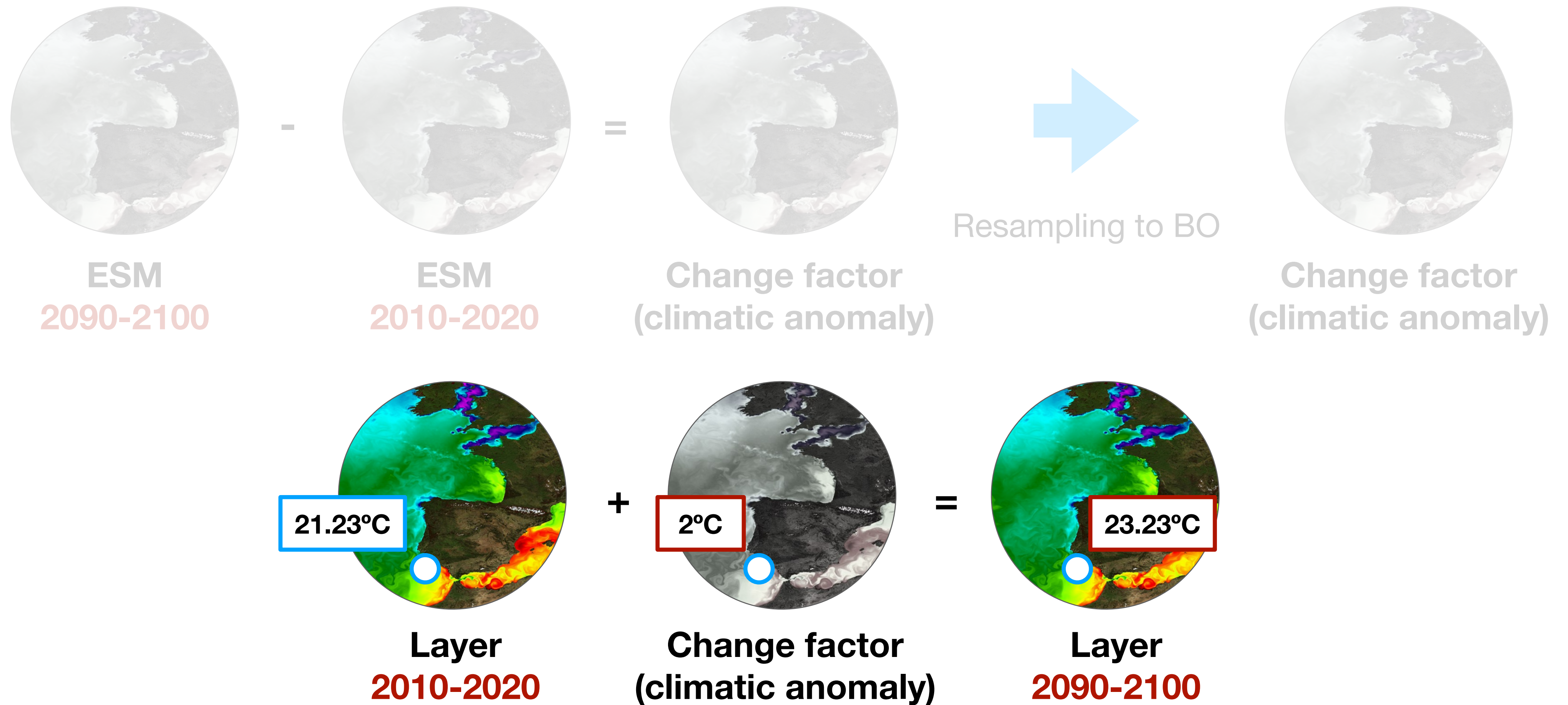
layer 2: Global maximum sea surface temperature of 2090-2100



For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

layer 2: Global maximum sea surface temperature of 2090-2100



For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

layer 2: Global maximum sea surface temperature of 2090-2100

The layers are provided under the FAIR principle at:

(1) An ERDDAP instance at the Flanders Marine Institute Data Centre **

<http://erddap.bio-oracle.org>

(2) Bio-ORACLE website

<http://www.bio-oracle.org>

(3) **R and Python packages** for programmatic data access and easy integration in available SDM workflows.

Code used to generate the layers available in the permanent GitHub repository at **<https://github.com/bio-oracle/data-layers>**.